Candidate Seat Number: ____________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Science in Information Technology (M.Sc. IT)
Semester-II University Examination May- 2018
MS 212.01 Software Engineering and Quality Assurance

Date: 09.05.2018, Wednesday   Time: 10.00 a.m. to 01.00 p.m.   Maximum Marks: 70

Instructions:
1. **Attempt both sections in separate answer books.**
2. **Figure to right indicates marks.**
3. **Make suitable assumption when necessary.**

SECTION-1

Q.1 [A] Do as directed **[05]**
1. In incremental model the whole requirement is divided in to number of builds. **(True/False).**
2. List the values of extreme programming.
3. **________** is the discipline of defining and achieving targets while optimizing the use of resources over the course of a project.
4. SRS stands for Software Requirement Simulation. **(True/False).**
5. Metric is a quantitative measure of the degree to which a system acquires a given attribute. **(True/False).**

[B] Answer the following in brief. **[06]**
1. List and Explain types of Software.
2. List and explain categories of measurement.

Q.2 [A] Explain the principles of Agile software development. **[04]**
[B] List and Explain 4Ps of management spectrum. **[04]**
[C] State the difference between Functional requirement vs. Nonfunctional requirement **[04]**

OR

Q.2 [A] List and explain activities of extreme programming. **[04]**
[B] List and Explain principles of project scheduling. **[04]**
[C] State the difference between Process indicators vs. Project indicator. **[04]**

Q.3[A] Write a note on Waterfall model. **[06]**
[B] Explain types of COCOMO. **[06]**

OR

Q.3 [A] Write a note on Spiral Model. **[06]**
[B] Explain characteristics of function oriented metrics with example. **[06]**

[P.T.O]

SECTION-II

Q.4 [A] Do as directed **[05]**

1. Define performance testing.
2. Code restructuring is not the process of design recovery. **(True/False).**
3. Quality refers to conformance of requirements. **(True/False).**
4. BPR stands for ________.
5. ________ assumes that mitigation efforts have failed and that the risk has become a reality**.**

[B] Answer the following in brief. **[06]**

1. Explain Boundary value analysis with suitable example.
2. Explain Reusability with example.

Q.5 [A] List and explain product revision factors. **[04]**

[B] List and explain risk identification check points. **[04]**

[C] State the difference between software validation vs. software verification. **[04]**

OR

Q.5 [A] List and explain product operation factors. **[04]**

[B] List and explain different categories of risk. **[04]**

[C] State the difference between Functional testing vs. Nonfunctional testing. **[04]**

Q.6[A] Write a note on Beta Testing. **[06]**

[B] Write a note on SQA goals, attributes and metrics. **[06]**

OR

Q.6 [A] Write a note on principles of software testing. **[06]**

[B] Write a note on elements of SQA **[06]**
